

ROBOT PLATFORM

Rosie

A six-axis articulated robot platform powered by AMR's GradientOS, the open-source industrial robotics controller.

OPEN SOURCE ★ NO FEES ★ EXPANDABLE IO ★ HOLLOW ROUTING

OPEN CONTROLLER

Full industrial robotics controller, open source, with no hidden software fees.

EASY INTEGRATION

Open GradientOS stack reduces lock-in and keeps functionality accessible.

EXPANDABLE IO

Ready for cell devices, tooling, safety signals, and end-effector integration.

HOLLOW CABLE PATHS

62 mm shoulder and 33 mm wrist pass-throughs for cleaner cable routing.



Payload

15 kg

Nominal handling capacity for Rosie.

Horizontal Reach

1,509 mm

Wide base-to-tool coverage for benches, fixtures, and compact automation cells.

Vertical Reach

1,638 mm

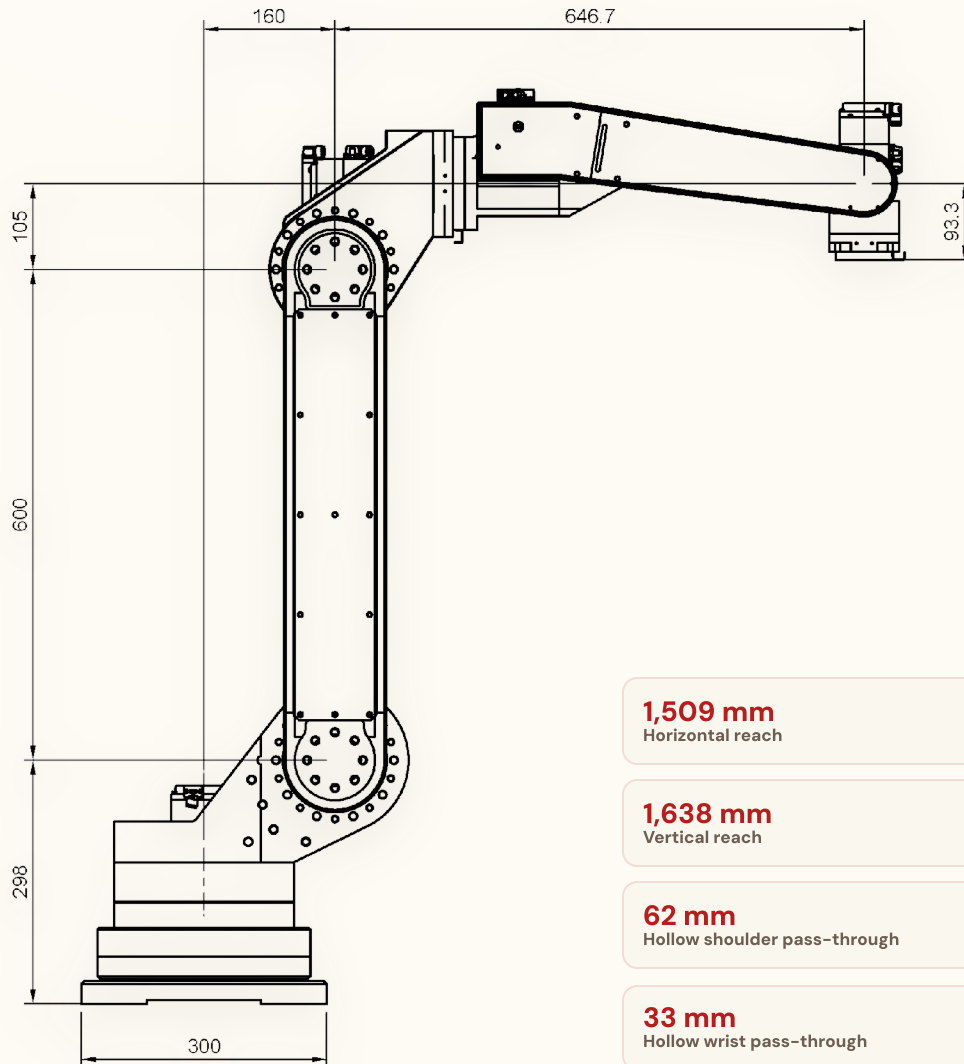
Above-base vertical reach; total envelope span is a separate sweep value.

Repeatability

0.08 mm

Repeatable positioning for fixture work, process validation, and automated cell routines.

Robot Dimensions



1,509 mm
Horizontal reach

1,638 mm
Vertical reach

62 mm
Hollow shoulder pass-through

33 mm
Hollow wrist pass-through

Axis Specifications

AXIS	ROLE	MOTION RANGE	RATED SPEED	PEAK SPEED
J1	Base	+/- 361.0 deg	180 deg/s	360 deg/s
J2	Shoulder	+/- 108.9 deg	180 deg/s	360 deg/s
J3	Elbow	-240.6 / +87.7 deg	180 deg/s	360 deg/s
J4	Wrist roll	+/- 361.0 deg	1,000 deg/s	2,000 deg/s
J5	Wrist pitch	+/- 361.0 deg	1,333 deg/s	2,667 deg/s
J6	Tool roll	+/- 573.0 deg	1,800 deg/s	3,600 deg/s

Open Controller

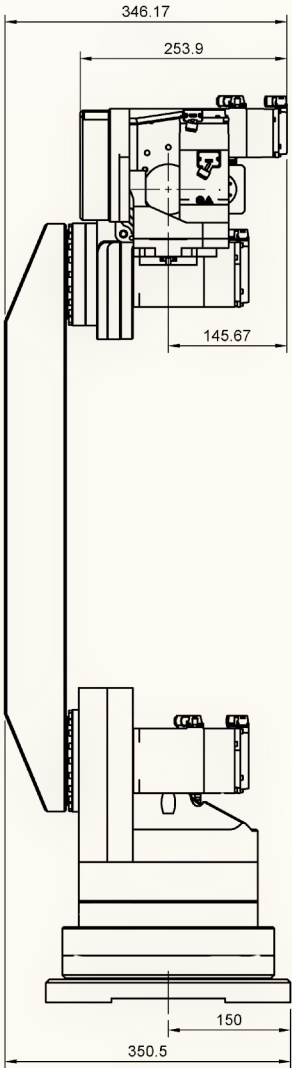
GradientOS is an open-source full industrial robotics controller, built to make integration simple without hidden software fees or artificially limited functionality.

Expandable IO supports cell devices, tooling, safety signals, and end-effector integration.

Controller Defaults

Robot ID	gradient-05	Gradient05Config
Physical actuators	6 EtherCAT axes	IDs 0-5
Profile velocity	0.1 m/s	default
Profile acceleration	0.05 m/s^2	default
Cartesian jog caps	0.2 m/s linear	180 deg/s angular
Encoder model	17-bit motor-side	absolute count model
Power	200-240 VAC 3PH	35A
Integration	Expandable IO	Open-source controller

Arm Envelope



346.17 mm
Wrist tool envelope

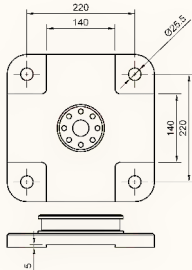
253.9 mm
Wrist housing span

145.67 mm
Tool-side offset

350.5 mm
Mount plate width

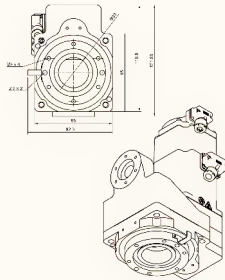
150 mm
Base column width

Base Mount



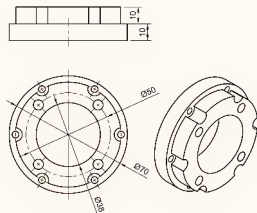
Mount plate	220 x 220 mm
Bolt circle	140 mm PCD
Anchor bore	Ø25.5 mm
Plate thickness	5 mm

Wrist Flange



Hollow bore	Ø33 mm
Bolt pattern	85 x 85 mm
Threaded mounts	M5 x 6
Flange width	92.3 mm

ISO50 Tool Adapter



Outer diameter	Ø70 mm
Bolt circle	Ø50 mm
Through bore	Ø38 mm
Stack height	10 mm